

Homework 13 Solutions

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Problem 30.1

Since the definition of a probability function requires $P(1) + P(2) + P(3) + P(4) = 1$, then $0.1 + 0.1 + 0.2 + x = 1 \Rightarrow x = 0.6$.

Problem 30.2

Since the definition of a probability function requires $P(1) + P(2) + P(3) + P(4) = 1$, then $x + 2x + 3x + 4x = 1 \Rightarrow 10x = 1 \Rightarrow x = 0.1$.

Problem 30.3

The equations $x + y = z$ and $z + y = x$ are given, and we know that $x + y + z = 1$ since P is a probability function. Then $x + y + z = 1 \Rightarrow z + z = 1 \Rightarrow z = 0.5$. Then $x + y = 0.5 \Rightarrow (y + z) + y = 0.5 \Rightarrow (0.5 + y) + y = 0.5 \Rightarrow 2y = 0 \Rightarrow y = 0$. Then $x = 0.5$.

Problem 30.4

$P(1) + P(2) + P(3) + P(4) = \frac{c}{1} + \frac{c}{4} + \frac{c}{9} + \frac{c}{16} = 1 \Rightarrow \frac{144c + 36c + 16c + 9c}{144} = 1 \Rightarrow 144c + 36c + 16c + 9c = 144 \Rightarrow 205c = 144 \Rightarrow c = \frac{144}{205}$

Problem 30.5

Proof. If (S, P_1) and (S, P_2) are both sample spaces, it is not possible that $\forall x \in S, P_1(x) < P_2(x)$.

1. Let (S, P_1) and (S, P_2) both be sample spaces using set S .
2. Assume by way of contradiction that $\forall x \in S, P_1(x) < P_2(x)$.
3. Then $\sum_{x \in S} P_1(x) < \sum_{x \in S} P_2(x)$.
4. Since P_1 and P_2 are both probability functions then $\sum_{x \in S} P_1(x) = 1$ and $\sum_{x \in S} P_2(x) = 1$.
5. Then $1 < 1 \Rightarrow \Leftarrow$.
6. Therefore it is not possible that $\forall x \in S, P_1(x) < P_2(x)$. □

Problem 30.10

$$\begin{aligned} P(1) &= \frac{\pi}{16\pi} = \frac{1}{16} \\ P(2) &= \frac{3\pi}{16\pi} = \frac{3}{16} \\ P(3) &= \frac{5\pi}{16\pi} = \frac{5}{16} \\ P(4) &= \frac{7\pi}{16\pi} = \frac{7}{16} \end{aligned}$$